

Ivan Evdokimov, Ph.D.

Belfast, UK | ivevdm@gmail.com | linkedin.com/in/ivan-evdokimov | github.com/ivan020

Professional Summary

Python Developer and Software Engineer with 6+ years of Python experience across enterprise-scale production systems and doctoral research in computational finance. Proven track record delivering end-to-end MLOps pipelines, LLM-powered applications, and FastAPI-based services in financial services environments. Experienced with MCP, RAG architecture, CI/CD, and AWS infrastructure, and comfortable collaborating across onshore and offshore teams to ship reliable, high-quality software.

Experience

Software Engineer – GenAI, University of Essex — Colchester, UK Nov 2025 — Present

- Designed and built enterprise-scale FastAPI services powering AI-driven development workflows, with automated deployment via GitHub Actions and Vercel achieving 90%+ code generation rate and weekly release.
- Designed automated testing and QA frameworks for AI-generated code, reducing manual review time by 70% while maintaining production stability and security standards.
- Built end-to-end MLOps infrastructure using Python and FastAPI for a multi-tenant SaaS platform serving global B2B clients, accelerating feature delivery from monthly to weekly releases.

Software Engineer – GenAI, UK Data Service — Colchester, UK Jan 2023 — Oct 2025

- Built and deployed FastAPI-based LLM and ML services (PyTorch, scikit-learn, Hugging Face, vLLM, Ollama) for automated classification, achieving 95%+ accuracy on sensitive datasets.
- Implemented RAG-style architectures for metadata classification, combining vector search with domain-specific language models to improve retrieval accuracy.
- Designed serverless ML infrastructure on AWS (Lambda, Step Functions, EFS) with automated retraining and monitoring pipelines, reducing manual workload by 50%.
- Collaborated with onshore and non-technical stakeholders to gather requirements and translate business needs into scalable technical solutions.
- Optimised statistical algorithms by migrating from R to C++ using bitmask techniques, achieving 11x performance improvement enabling real-time processing of 500k+ record datasets.

Software Engineer & Instructor, University of Essex — Colchester, UK Oct 2021 — Dec 2022

- Migrated quantitative macroeconomic models from MATLAB to C++, reducing simulation runtime by 8x for cross-university research collaborations in the UK and USA.
- Taught Data Structures, C/C++, and Machine Learning to 100+ undergraduate and postgraduate students, developing strong written and verbal communication skills.

Education

University of Essex, PhD in Computational Finance Oct 2021 — May 2025

- Built high-performance C++ data pipelines to source, process, and clean large-scale market and fundamental financial datasets from Bloomberg, feeding AI-driven model training workflows.
- Engineered a PyTorch-based transfer learning framework for financial return prediction and portfolio optimisation, producing open-source tooling adopted externally.
- Applied ML and Bayesian approaches to develop and back-test trading and investment strategies using fundamental financial data — providing direct grounding in financial services domain.

University of Essex, MSc in Financial Econometrics Oct 2020 — Sep 2021

- Built a C++ agent-based financial simulation modelling macroeconomic dynamics under negative interest rates.

Projects

FinBot: Chat-Based Trading Simulation github.com/ivan020/FinBot

- Built a real-time trading simulation system in Go with live order entry, a PostgreSQL (Supabase) backend recording transactions, and real-time P&L tracking — analogous to order management system design in financial services.
- Developed a JavaScript UI for real-time portfolio visualisation with interactive graphs and tables.

Chat-LLM github.com/ivan020/ChatLLM

- Built a self-hosted Ollama MCP server on a Raspberry Pi 5, implementing custom tools for AI-driven story generation about Twitch viewers and automated anime-style image generation posted live to stream.

MTG Card Reader <https://mtg-front-end-nine.vercel.app/>

- Built a Golang REST API with Mistral LLM integration and PostgreSQL backend, enabling automatic card recognition from photos with real-time price lookup.
- Developed a React frontend for card cataloguing, filtering, and portfolio management — reaching 50 monthly active users.